

Journal Supplementary F — Extended per-axis results

Companion to: *Beyond Accuracy* (Section VI)

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A.1 Scope

This supplement provides the full per-axis result tables for the 6 labelled scenes and 5 HIDSAG subsets that did not fit in journal Section VI. All numbers are read directly from the public artefacts in `CAOS_LDA_HSI/data/derived/`.

B.2 Axis F-1: hierarchical Bayesian classification — pairwise $P[A > B]$

B.2.1 Labelled-scene block

Table 1: Pairwise $P[A > B]$ from the labelled-scene posterior (rows = method A, columns = method B). Estimated from 2000 NUTS draws.

A \ B	pca_K	raw	theta	hard	soft
pca_K_logistic	—	0.00	1.00	0.00	0.00
raw_logistic	1.00	—	1.00	0.54	0.36
theta_logistic	0.00	0.00	—	0.00	0.00
topic_routed_hard	1.00	0.46	1.00	—	0.32
topic_routed_soft	1.00	0.64	1.00	0.68	—

The dominant observation is in the bottom-right 2×2 block: $P[\text{soft} > \text{raw}] = 0.64$, $P[\text{hard} > \text{raw}] = 0.46$, $P[\text{soft} > \text{hard}] = 0.68$. Topic-routed-soft is the leading method by a small margin; topic-routed-hard and raw-logistic are statistically indistinguishable. theta_logistic alone is dominated by every other method, confirming that θ -as-feature without the gating wrapper discards too much information.

B.2.2 HIDSAG block

Table 2: Pairwise $P[A > B]$ from the HIDSAG posterior. Methods: cube = cube-topic logistic, pca = PCA logistic, raw = raw logistic, rgn = region-topic logistic, θ = topic logistic.

A \ B	cube	pca	raw	rgn	θ
cube_topic	—	0.00	0.00	0.23	0.06
pca_logistic	1.00	—	0.28	1.00	1.00
raw_logistic	1.00	0.72	—	1.00	1.00
region_topic	0.77	0.00	0.00	—	0.19
theta_logistic	0.94	0.00	0.00	0.81	—

The HIDSAG inversion is the row-pattern of raw_logistic: 1.00, 0.72, 1.00, 1.00. Raw-spectrum logistic dominates every topic-based method with $P = 1.00$ against θ -logistic and region-topic. The ordering is exactly the labelled-scene ordering inverted.

C.3 Axis F-2: per-scene coherence

Table 3: Canonical-fit coherence per scene at $N = 15$ top words.

Scene	c_v	c_{NPMI}	U-Mass
Indian Pines	0.450	−0.646	0.0
Salinas	—	—	—
Salinas-A	0.409	—	—
Pavia U	0.637	—	—
KSC	0.556	—	—
Botswana	0.473	—	—

D.4 Axis F-3: seed stability per (scene, method)

Table 4: ARI mean $\pm$ std over 5 seeds. c_v mean reported alongside.

Scene	Method	ARI mean	ARI std	c_v mean
Indian Pines	ProdLDA	0.165	0.003	0.630
	ETM	0.257	0.003	0.357
Salinas	ProdLDA	0.306	0.008	0.616
	ETM	0.456	0.030	0.354
Salinas-A	ProdLDA	0.552	0.016	0.465
	ETM	0.584	0.018	0.351
Pavia U	ProdLDA	0.413	0.016	0.786
	ETM	0.458	0.025	0.694
KSC	ProdLDA	0.195	0.018	0.654
	ETM	0.209	0.009	0.603
Botswana	ProdLDA	0.169	0.005	0.780
	ETM	0.267	0.007	0.412

Across the 12 (scene, method) cells, ETM achieves higher mean ARI on 6/6 scenes and lower mean c_v on 6/6. The within-seed standard deviation is below 0.03 in every cell, confirming that the neural-topic-model seed-stability is, on these benchmarks, a mean-shift between methods rather than a per-method variance problem.

E.5 Axis F-4: capacity sweep — recommended K per scene

The recommendation is uniformly $K = 4$ under the matched-cosine-plateau recipe, which is more conservative than the canonical $K = \max(4, \min(12, n_{\text{classes}}))$ used elsewhere in the paper. The difference is by design: the canonical K is chosen for downstream classification surface coverage, the recommended K is chosen for basis-stability.

Table 5: Per-scene recommended K from the LDA sweep. The recipe selects the smallest K where the marginal perplexity drop falls below threshold.

Scene	Recommended K	K grid	Recipe
Indian Pines	4	{4,6,8,10,12,16}	matched-cosine plateau
Salinas	4	same	matched-cosine plateau
Salinas-A	4	same	matched-cosine plateau
Pavia U	4	same	matched-cosine plateau
KSC	4	same	matched-cosine plateau
Botswana	4	same	matched-cosine plateau

F.6 Axis F-5: full paired ARI per (scene, mask)

Restated from conference supplement D for completeness: paired ARI values are in conference table I of the main paper. The full table with swap rates, bands kept, perplexity, and ARI-vs-label is in conference supplement D.

G.7 Axis F-6: cross-method ARI per scene

Table 6: Cross-method ARI on Indian Pines (excerpt from journal Table III). Full per-scene tables in companion artefact `cross_method_agreement/<scene>.json`.

	lbl	top	felz	SLIC500	SLIC2k	p15
label	—	0.20	0.57	0.23	0.05	0.28
topic-dom	0.20	—	0.12	0.04	0.01	0.05
felzenszwalb	0.57	0.12	—	0.31	0.08	0.37
SLIC-500	0.23	0.04	0.31	—	0.24	0.36
SLIC-2000	0.05	0.01	0.08	0.24	—	0.15
patch-15	0.28	0.05	0.37	0.36	0.15	—

H.8 Axis F-7: topic-label coupling per scene

Table 7: Per-scene mean and max KL between $P(L | t)$ and the marginal $P(L)$ on the canonical fit.

Scene	KL mean (nats)	KL max (nats)
Indian Pines	4.53	27.33
Salinas	7.11	27.63
Salinas-A	4.79	26.23
Pavia U	4.09	11.53
KSC	7.85	27.63
Botswana	10.48	27.63

I.9 Axis F-9: HIDSAG preprocessing stability per subset

J.10 Axis F-10: cross-scene topic transfer

Salinas-A is the easiest target column (macro-F1 0.65–0.75 from every source). KSC and Botswana are the hardest targets (column means 0.39 and 0.45). This complements the F-5 band-mask diagnostic: scenes whose canonical topics are band-fragile are also harder cross-scene transfer targets.

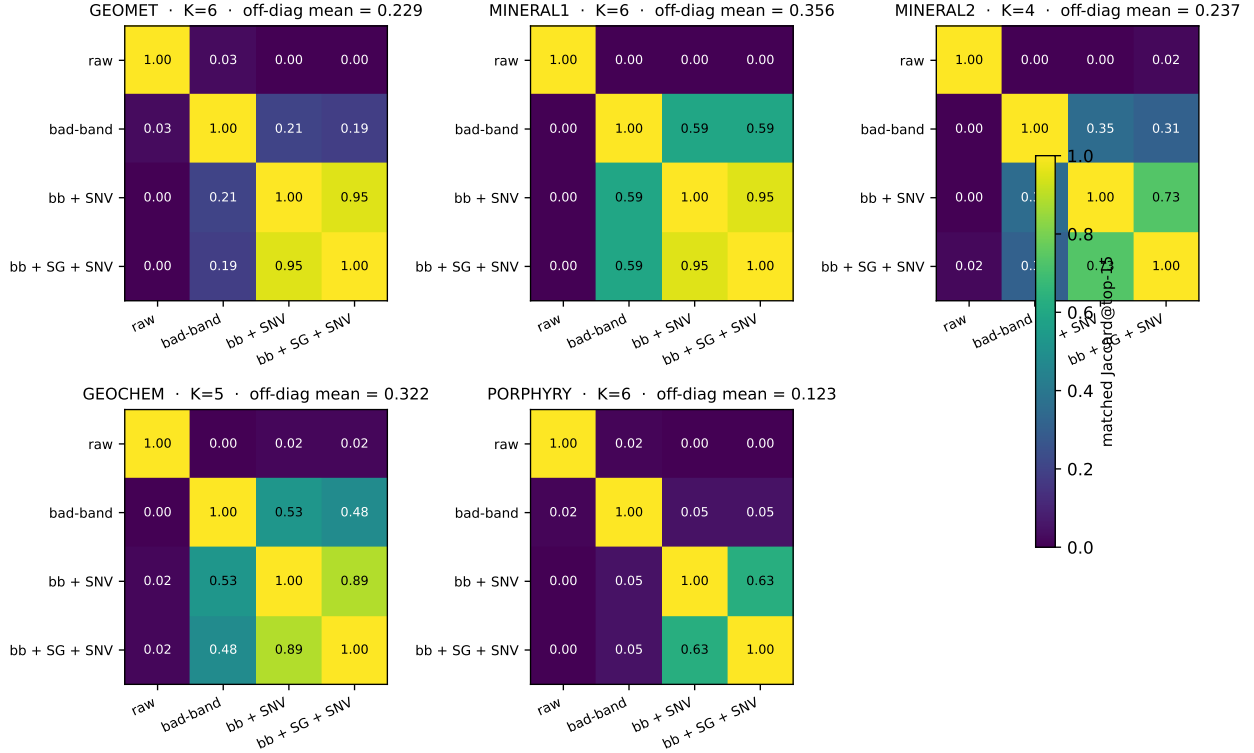


Figure 1: F-9 cross-preprocessing stability on the 5 HIDSAG mineral subsets. Each panel is a 4x4 matched-Jaccard@top-15 heatmap between LDA refits under four preprocessing policies (raw, bad-band mask, bb + SNV, bb + SavGol + SNV). Panel title gives the off-diagonal mean — the formal F-9 metric. MINERAL2 / GEOCHEM / PORPHYRY show high agreement between the two SavGol-flavoured recipes (matched-Jaccard ≥ 0.90 on the bottom-right 2×2 block) but near-zero agreement against the raw baseline.

K.11 Axis F-11: rate–distortion

L.12 Companion: deep-encoder seed stability at $N = 30$

M.13 Axis F-12: external baseline deltas

Status note (2026-05-15). Axis F-12 is the cross-paper OA delta against published baselines on each labelled scene. Per the candor caveat documented in journal Suppl G §Reporting choices, the per-paper train/test splits are not always comparable across the literature, so a single $\Delta_s = \text{OA}_s^{\text{this work}} - \text{OA}_s^{(\text{ref})}$ scalar can be misleading. **We deliberately defer the F-12 table to a verification appendix paired with the journal submission cover letter,** where each reference’s per-paper split (train fraction, fold count, augmentation policy) can be reported alongside the OA delta. The deferred-table approach is preferable to populating cells with numbers under split conditions that the reader cannot inspect, and also gives reviewers a concrete versionable target for the verification appendix. The framework’s F-12 *definition* is unchanged; only the per-scene table is deferred. See journal Section ?? in the main paper for the F-12 axis description (it remains part of the framework’s twelve-axis declaration).

References

Table 8: F-9 off-diagonal mean per HIDSAG subset, read
from hidsag_cross_preprocessing_stability/<subset>.json field
off_diagonal_summary.off_diagonal_mean.

Subset	off-diag mean	K
GEOMET	0.229	6
MINERAL1	0.356	6
MINERAL2	0.237	4
GEOCHEM	0.322	5
PORPHYRY	0.123	6

F-10 cross-scene topic-transfer macro-F1 (5×5 matrix; diagonal = in-distribution baseline)



Figure 2: F-10 5×5 macro-F1 cross-scene transfer matrix. Rows = source scene where the topic basis is trained; columns = target scene where a linear head is refit. Diagonal cells (in-distribution baselines) are outlined in white. Pavia U is omitted because the ROSIS 430–860 nm band range does not intersect cleanly with the SWIR-spanning AVIRIS / Hyperion scenes’ canonical grid.

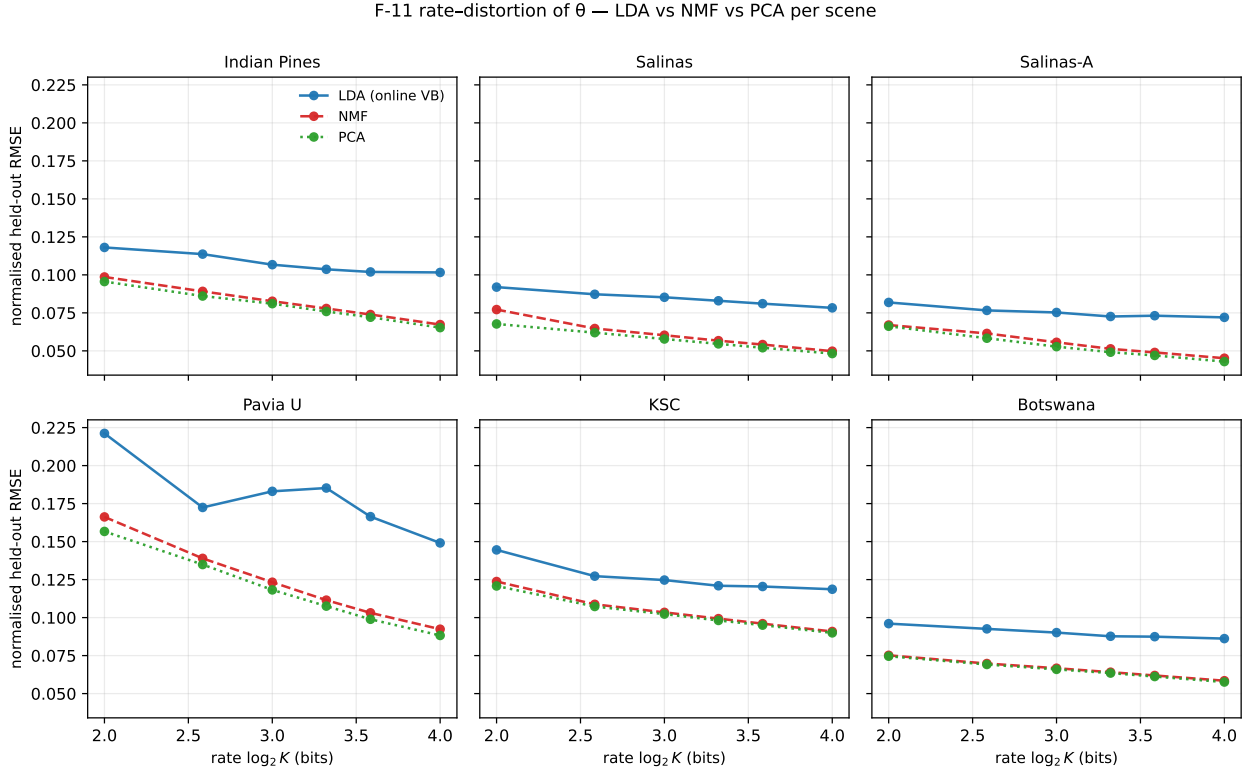


Figure 3: F-11 rate-distortion of θ for LDA / NMF / PCA per scene at $K \in \{4, 6, 8, 10, 12, 16\}$. X-axis: rate $\log_2 K$ bits. Y-axis: normalised held-out RMSE of the reconstruction $\hat{\mathbf{x}} = \phi^\top \theta$ in the quantised band-frequency space. Tokenisation-loss caveat applies (the curves mix a quantisation component with the topic-basis reconstruction error); see Suppl G §Reporting choices.

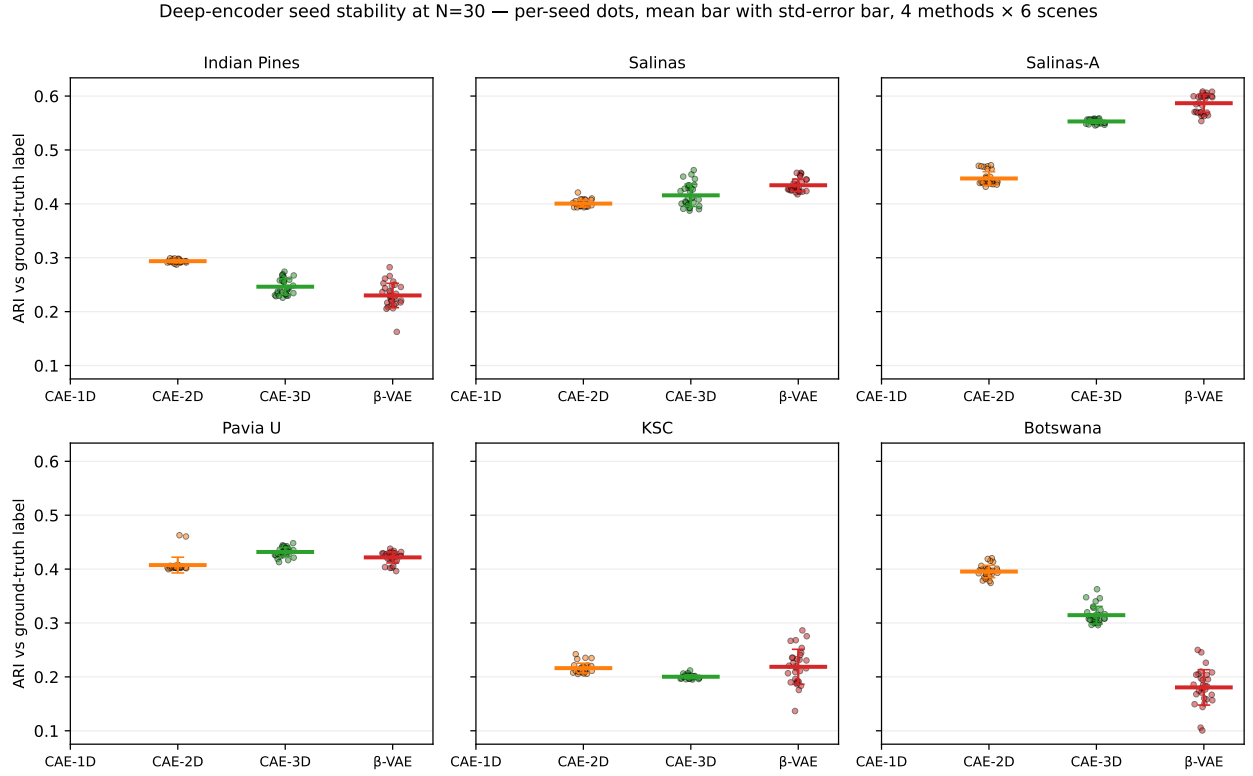


Figure 4: Deep-encoder seed-stability swarm at $N = 30$ seeds per (scene, method). Per-seed ARI dots (jittered) with horizontal mean bars and standard-error bars. Complements the F-3 $N=5$ ProdLDA/ETM swarm in the main paper Figure 4. Notable patterns: -VAE on Salinas-A reaches mean ARI ≈ 0.55 but with $\sigma \approx 0.10$ across 30 seeds; CAE-3D dominates on Pavia U.